Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The adjacent sides AB and AC of a triangle ABC are represented by the vectors $-2\hat{i}+3\hat{j}+2\hat{k}$ and $-4\hat{i}+5\hat{j}+2\hat{k}$ respectively. The area of the triangle ABC is

(A) 6 square units

(B) 5 square units

(C) 4 square units

(D) 3 square units

Q.2

Resolved part of vector $\vec{a}$ along vector $\vec{b}$ is $\vec{a}_1$ and that perpendicular to $\vec{b}$ is $\vec{a}_2$, then $\vec{a}_1 \times \vec{a}_2$ is equal to

(A) $\frac{(\vec{a}\times\vec{b})\cdot\vec{b}}{|\vec{b}|^2}$

(B) $\frac{(\vec{a}\cdot\vec{b})\vec{a}}{|\vec{a}|^2}$

(C) $\frac{(\vec{a}\cdot\vec{b})(\vec{b}\times\vec{a})}{|\vec{b}|^2}$

(D) $\frac{(\vec{a}\cdot\vec{b})(\vec{b}\times\vec{a})}{|\vec{b}\times\vec{a}|}$

Q.3

The components of a vector $\vec{a}$ along and perpendicular to a non-zero vector $\vec{b}$ are

(A) $\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{b}\&\vec{a}-\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{b}$

(B) $\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{a}|^2}\right)\vec{b}\&\vec{a}+\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{a}|^2}\right)\vec{b}$

(C) $\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{a}|^2}\right)\vec{a}-\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{a}$

(D) None of these

Q.4

In a right angle $\triangle ABC, \angle A=90^\circ$ and sides a, b, c are respectively, $5\ \mathrm{cm}, 4\ \mathrm{cm}$ and $3\ \mathrm{cm}$. If a force

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$\vec{F}$ has moments 0,9 and 16 in N cm. units respectively about vertices A, B and C, then magnitude of $\vec{F}$ is

(A) 9

(B) 4

(C) 5

(D) 3

Q.5

If the pth , q^{th} and r^{th} terms of a G.P. are positive numbers a, b and c respectively, then find the angle between the vectors

$\log a^2\hat{i} + \log b^2\hat{j} + \log c^2\hat{k}$ and

$(q-r)\hat{i} + (r-p)\hat{j} + (p-q)\hat{k}$

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{2}$

Q.6

What is the area of the parallelogram having diagonals $3\hat{i} + \hat{j} - 2\hat{k}$ and $\hat{i} - 3\hat{j} + 4\hat{k}$?

(A) $5\sqrt{5}$ square units

(B) $4\sqrt{5}$ square units

(C) $5\sqrt{3}$ square units

(D) $15\sqrt{2}$ square units

Q.7

A force $\vec{F} = 3\hat{i} + 4\hat{j} - 3\hat{k}$ is applied at the point P, whose position vector is $\vec{r} = \hat{2}i - 2\hat{j} - 3\hat{k}$. What is the magnitude of the moment of the force about the origin?

(A) 23 units

(B) 19 units

(C) 18 units

(D) 21 units

Q.8

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If $\vec{p}$ and $\overrightarrow{q}$ are non-collinear unit vectors and $\left|\overrightarrow{p}+\overrightarrow{q}\right|=\sqrt{3}$, then $(2\overrightarrow{p}-3\overrightarrow{q})\cdot(3\overrightarrow{p}+\overrightarrow{q})$ is equal to

(A) 0

(B) $\frac{1}{3}$

(C) $-\frac{1}{3}$

(D) $-\frac{1}{2}$

Q.9

A force $F=2i+j-k$ acts at a point A, whose position vector is $2i-j$. The moment of F about the origin is

(A) $i+2j-4k$

(B) $i-2j-4k$

(C) $i+2j+4k$

(D) $i-2j+4k$

Q.10

If $\vec{a}=2\hat{i}+2\hat{j}+3\hat{k}, \vec{b}=-\hat{i}+2\hat{j}+\hat{k}$ and $\vec{c}=3\hat{i}+\hat{j}$
are three vectors such that $\vec{a}+t\vec{b}$ is
perpendicular to $\vec{c}$, then what is t equal to ?

(A) 8

(B) 6

(C) 4

(D) 2

Q.11

A vector $\overrightarrow{\mathrm{a}}=(\mathrm{x},\mathrm{y},\mathrm{z})$ of length $2\sqrt{3}$ which makes equal angles with the vectors $\overrightarrow{\mathrm{b}}=(\mathrm{y},-2\mathrm{z},3\mathrm{x})$ and $\overrightarrow{\mathrm{c}}=(2\mathrm{z},3\mathrm{x},-\mathrm{y})$ and is
perpendicualr to $\overrightarrow{\mathrm{d}}=(1,-1,2)$ and makes an obtuse angle with y -axis is

(A) $(-2,2,2)$

(B) $(1,1,\sqrt{10})$

(C) $(2,-2,-2)$

(D) None of these

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Q.12

Force $\hat{i}+2\hat{j}-3\hat{k}, 2\hat{i}+3\hat{j}+4\hat{k}$ and $-\hat{i}-\hat{j}+\hat{k}$ are
acting at the point $\mathrm{P}(0,1,2)$. The moment of these forces about the point $\mathrm{A}(1,-2,0)$ is

(A) $2\hat{i}-6\hat{j}+10\hat{k}$

(B) $-2\hat{i}+6\hat{j}-10\hat{k}$

(C) $2\hat{i}+6\hat{j}-10\hat{k}$

(D) None of these

Q.13

The value of $'x'$ for which the angle between the vectors $\vec{a}=2x^2\hat{i}+4\hat{x}\hat{j}+\hat{k}$ and
$\overrightarrow{\mathrm{b}}=7\hat{\mathrm{i}}-2\hat{\mathrm{j}}+\mathrm{x}\hat{\mathrm{k}}$ is obtuse are

(A) $x<0$

(B) $x>\frac{1}{2}$

(C) $x<\frac{1}{2}$

(D) $0<x<\frac{1}{2}$

Q.14

If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are the position vectors of points $\mathrm{A}\ \mathrm{B}, \mathrm{C}$ and D respectively such that $(\vec{a}-\vec{d})\cdot(\vec{b}-\vec{c})=(\vec{b}-\vec{d})\cdot(\vec{c}-\vec{a})=0$ then D is the

(A) centroid of $\triangle\mathrm{ABC}$

(B) circumcentre of $\triangle\mathrm{ABC}$

(C) orthocentre of $\triangle\mathrm{ABC}$

(D) None of these

Q.15

The vectors $\hat{i}-2x\hat{j}-3y\hat{k}$ and $\hat{i}+3x\hat{j}+2y\hat{k}$ are orthogonal to each other. Then the locus of the point (x,y) is

(A) hyperbola

(B) ellipse

(C) parabola

(D) circle

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Q.16

If $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{k}$ then $|\vec{c}| \cdot \vec{a}$

(A) $2\sqrt{5}\hat{i} + 2\sqrt{5}\hat{j} + \sqrt{5}\hat{k}$

(B) $2\sqrt{5}\hat{i} - 2\sqrt{5}\hat{j} + \sqrt{5}\hat{k}$

(C) $\sqrt{5}\hat{i} + 2\sqrt{5}\hat{j} + \sqrt{5}\hat{k}$

(D) $\sqrt{5}\hat{i} + \sqrt{5}\hat{j} + \sqrt{5}\hat{k}$

Q.17

If $a^2 + b^2 + c^2 = 1$ where $a, b, c \in R$, then the maximum value of $(4a - 3b)^2 + (5b - 4c)^2 + (3c - 5a)^2$ is

(A) 25

(B) 50

(C) 144

(D) None of these

Q.18

The postion vectors of the vertices of a quadrilateral ABCD are $\mathrm{a}, \mathrm{b}, \mathrm{c}$ and d respectively. Area of the quadrilateral formed by joining the middle points of its sides is

(A) $\frac{1}{4}|\mathrm{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{d} + \mathbf{d} \times \mathbf{a}|$

(B) $\frac{1}{4}|\ \mathrm{b} \times \mathrm{c} + \mathrm{c} \times \mathrm{d} + \mathrm{axd} + \mathrm{b} \times \mathrm{a}|$

(C) $\frac{1}{4}|\mathrm{a} \times \mathrm{b} + \mathrm{b} \times \mathrm{c} + \mathrm{c} \times \mathrm{d} + \mathrm{d} \times \mathrm{a}|$

(D) $\frac{1}{4}\ |\ \mathrm{b} \times \mathrm{c} + \mathrm{c} \times \mathrm{d} + \mathrm{d} \times \mathrm{b}$

Q.19

The component of vetor $i + j + k$ along vector, $i + 2j + 3k$ is

(A) $(3/7)(\mathrm{i} + 2\mathrm{j} + 3\mathrm{k})$

(B) $(2/7)(i + 2j + 3k)$

(C) $(1/7)(\mathrm{i} + 2\mathrm{j} + 3\mathrm{k})$

(D) $(4/7)(i + 2j + 3k)$

Q.20

Area of the parallelogram whose diagonals are $\vec{a} = 3\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 3\hat{j} + 4k$ is

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(A) $10\sqrt{3}$

(B) $\frac{3}{2}\sqrt{30}$

(C) 8

(D) 4

Answers & Solutions

Q.1 Answer:

3 square units

Solution:

$$\text{Area of } \Delta ABC = \frac{1}{2}(\overrightarrow{AB} \times \overrightarrow{AC})$$
$$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 3 & 2 \\ -4 & 5 & 2 \end{vmatrix}$$
$$= \frac{1}{2}[\hat{i}(6-10) - \hat{j}(-4+8) + \hat{k}(-10+12)]$$
$$= \frac{1}{2}[-4\hat{i} - 4\hat{j} + 2\hat{k}]$$
$$= \frac{1}{2}\sqrt{16+16+4} = \frac{1}{2}\sqrt{36}$$
$$= \frac{1}{2} \times 6$$
$$= 3 \text{ square units}$$

Q.2 Answer:

$\frac{(\vec{a}\cdot\vec{b})(\vec{b}\times\vec{a})}{|\vec{b}|^2}$

Solution:

$$\vec{a}_1 = (\vec{a}\cdot\hat{b})\hat{b} = \frac{(\vec{a}\cdot\vec{b})\vec{b}}{|\vec{b}|^2}$$
$$\Rightarrow \vec{a}_2 = \vec{a} - \vec{a}_1 = \vec{a} - \frac{(\vec{a}\cdot\vec{b})\vec{b}}{|\vec{b}|^2}$$
$$\text{Thus, } \vec{a}_1 \times \vec{a}_2 = \frac{(\vec{a}\cdot\vec{b})\vec{b}}{|\vec{b}|^2} \times \left(\vec{a} - \frac{(\vec{a}\cdot\vec{b})\vec{b}}{|\vec{b}|^2}\right)$$
$$= \frac{(\vec{a}\cdot\vec{b})(\vec{b}\times\vec{a})}{|\vec{b}|^2}$$

Q.3 Answer:

$\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{b} \& \vec{a} - \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{b}$

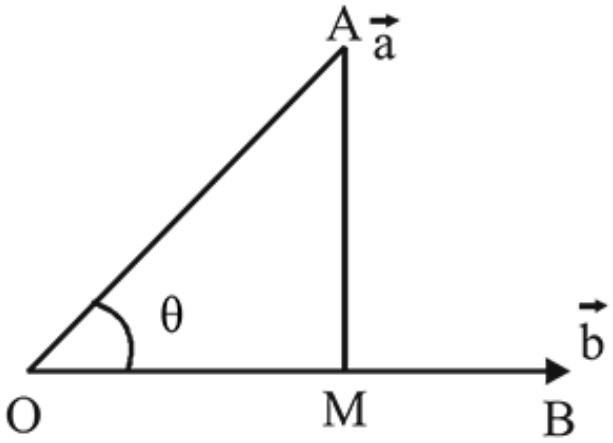
Solution:

$\overrightarrow{OM}$ = **component of** $\vec{a}$ **along** $\vec{b}$

$\overrightarrow{MA}$ = **component of** $\vec{a}$

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perpendicular to $\vec{b}$



$\Delta OMA \Rightarrow \cos\theta = \frac{OM}{OA}$

$\Rightarrow OM = |\overrightarrow{OM}| = |\overrightarrow{OA}|\cos\theta = |\overrightarrow{a}|\cos\theta$

$\because \vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta = |\vec{b}|(OM)$

$\therefore \overrightarrow{OM} = |\overrightarrow{OM}|\hat{b} = \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}\right)\frac{\vec{b}}{|\vec{b}|} = \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{b}$

$\overrightarrow{OM} + \overrightarrow{MA} = \overrightarrow{OA}$

$\therefore \overrightarrow{MA} = \overrightarrow{OA} - \overrightarrow{OM}$

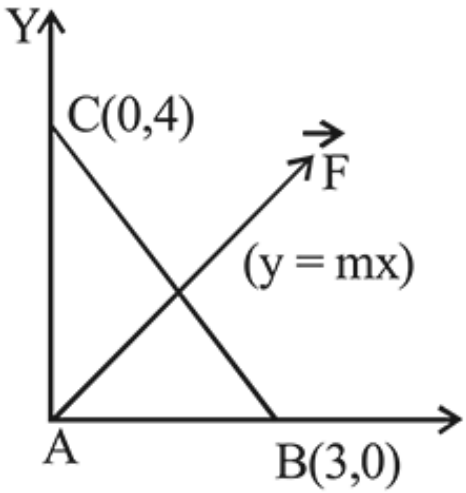
$= \vec{a} - \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{b}$

Q.4 Answer:

5

Solution:

Since, the moment about A is zero, hence $\vec{F}$ passes through A. Taking A as origin. Let the line of action of force $\vec{F}$ be $y = mx$. (see figure) Moment about $B = \frac{3m}{\sqrt{1+m^2}}|\vec{F}| = 9 \ldots..(1)$



Moment about

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$C = \frac{4}{\sqrt{1+m^2}}|\vec{F}| = 16 \ldots..(2)$

Dividing (1) by (2), we get : $m = \frac{3}{4} \Rightarrow |\vec{F}| = 5N.$

Q.5 Answer:

$\frac{\pi}{2}$

Solution:

Let A be the first term and x the common ratio of G.P.

So, $a = Ax^{p-1} \Rightarrow \log a = \log A + (p-1)\log x$

Similarly, $\log \mathrm{b} = \log \mathrm{A} + (\mathrm{q}-1)\log \mathrm{x}$

and $\log c = \log A + (r-1)\log x$

If $\vec{\alpha} = \log \mathrm{a}^2\hat{\mathrm{i}} + \log \mathrm{b}^2\hat{\mathrm{j}} + \log \mathrm{c}^2\hat{\mathrm{k}}$

and $\vec{\beta} = (\mathrm{q}-\mathrm{r})\hat{\mathrm{i}} + (\mathrm{r}-\mathrm{p})\hat{\mathrm{j}} + (\mathrm{p}-\mathrm{q})\hat{\mathrm{k}}$ **then**

$\vec{\alpha}\cdot\vec{\beta} = 2[\log \mathrm{a}(\mathrm{q}-\mathrm{r}) + \log \mathrm{b}(\mathrm{r}-\mathrm{p}) + \log \mathrm{c}(\mathrm{p}-\mathrm{q})]$

$= 2[(\mathrm{q}-\mathrm{r})\{\log \mathrm{A} + (\mathrm{p}-1)\log \mathrm{x}\}$

$+(\mathrm{r}-\mathrm{p})\{\log \mathrm{A} + (\mathrm{q}-1)\log \mathrm{x}\}$

$+(\mathrm{p}-\mathrm{q})\{\log \mathrm{A} + (\mathrm{r}-1)\log \mathrm{x}\}]$

$+(\mathrm{qp}-\mathrm{pr}-\mathrm{p}+\mathrm{r}+\mathrm{qr}-\mathrm{pq}$

$-2[(\mathrm{q}-\mathrm{r}+\mathrm{r}-\mathrm{p}+\mathrm{p}-\mathrm{q})\log \mathrm{A}$

$-\mathrm{r}+\mathrm{p}+\mathrm{pr}-\mathrm{qr}-\mathrm{p}+\mathrm{q})\log \mathrm{x}] = 0$

Hence, the angle between $\vec{\alpha}$ and $\vec{\beta}$ is $\frac{\pi}{2}$.

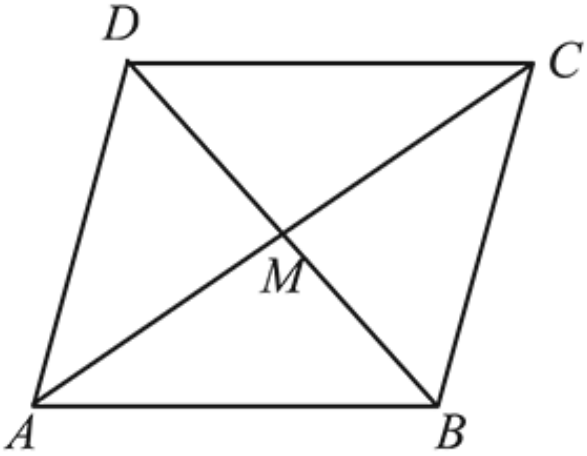
Q.6 Answer:

$5\sqrt{3}$ square units

Solution:

Diagonal $d_1, \overrightarrow{AC} = 3i + j - 2k$

Diagonal $d_2, \overrightarrow{BD} = i - 3j + 4k$



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Area of parallelogram is $\frac{1}{2}\left|\overrightarrow{d_1} \times \overrightarrow{d_2}\right|$ **Hence area** $= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 1 & -3 & 4 \end{vmatrix}$

$= \frac{1}{2} \mid [\hat{i}(4-6) - \hat{j}(12+2) + \hat{k}(-9-1)$

$= \frac{1}{2}|-2\hat{i} - 14\hat{j} - 10\hat{k}|$

$= \frac{1}{2}\sqrt{4+196+100}$

$= \frac{10\sqrt{3}}{2} = 5\sqrt{3}$ **square units**

Q.7 Answer:

23 units

Solution:

Moment of force, $\mathrm{m} = \overrightarrow{\mathrm{r}} \times \overrightarrow{\mathrm{F}}$

$m = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & -3 \\ 3 & 4 & -3 \end{vmatrix}$

$= \hat{\mathrm{i}}(6+12) - \hat{\mathrm{j}}(-6+9) + \hat{\mathrm{k}}(8+6)$

$= 18\hat{\mathrm{i}} - 3\hat{\mathrm{j}} + 14\hat{\mathrm{k}}$

$= \sqrt{(18)^2 + (-3)^2 + (14)^2}$

$= \sqrt{529} = 23$ **units.**

Q.8 Answer:

$-\frac{1}{2}$

Solution:

$|\vec{p} + \vec{q}| = \sqrt{3} \Rightarrow \vec{p}^2 + q^2 + 2\vec{p}\vec{q} = 3$

Since p **and** q **are unit vectors So,** $1 + 1 + 2\mathrm{pq} = 3$

$\Rightarrow 2\mathrm{pq} = 1 \Rightarrow \mathrm{pq} = \frac{1}{2}$

$(2\vec{p} - 3\vec{q})(3\vec{p} + \vec{q}) = 6\vec{p}^2 + 2\vec{p}\vec{q} - 9\vec{q}\vec{p} - 3\vec{q}^2 = \frac{-1}{2}$

Q.9 Answer:

$i + 2j + 4k$

Solution:

Force $\left(\overrightarrow{\mathrm{F}}\right) = 2i + j - k$ **and its position vector of** $\mathrm{A} = 2i - j$**. We know that the position vector of force about origin** $(r) = (2i - j) - (0i + 0j + 0k)$ **or** $r = 2i - j$.

Therefore, moment of the force about origin $= r \times \vec{F} = \begin{vmatrix} i & j & k \\ 2 & -1 & 0 \\ 2 & 1 & -1 \end{vmatrix} = i + 2j + 4k$

Q.10 Answer:

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Solution:

$\vec{a}+t\vec{b}=(2-t)\hat{i}+(2+2t)\hat{j}+(3+t)\hat{k}$

$(\vec{a}+t\vec{b})$ **and** $\vec{c}$ **is perpendicular. Therefore,**

$(\vec{a}+t\vec{b})\cdot\vec{c}=0$

$3(2-t)+2+2t=0$

$6-3t+2t+2=0$

$\mathrm{t}=8$

Q.11 Answer:

$(2,-2,-2)$

Solution:

Since, $\vec{a}$ **is** $\perp$ **to** $\vec{d}$**, so** $x-y+2z=0$

Moreover, $|\overrightarrow{\mathrm{b}}|=|\overrightarrow{\mathrm{c}}|$**, so** $\overrightarrow{\mathrm{a}}\cdot\overrightarrow{\mathrm{b}}=\overrightarrow{\mathrm{a}}\cdot\overrightarrow{\mathrm{c}}$

as a makes equal angles with $\overrightarrow{\mathrm{b}}$ **and** $\overrightarrow{\mathrm{c}}$**. Thus** $xy-2yz+3xz=2xz+3xy-yz$

$\Rightarrow xz-2xy-yz=0$

Also, $x^2+y^2+z^2=12$

and $y<0$ **Put the value of y from eq. (1) in eq. (2), we get,** $x^2+2xz+z^2=0$**; so,** $x=-z$ **and** $y=z$ **Again put these values in eq. (3), we get** $z^2=4\Rightarrow z=\pm2$

But $y<0$ **and** $y=z$**. Hence,** $z=-2=y$ **and** $x=2$

Q.12 Answer:

$-2\hat{i}+6\hat{j}-10\hat{k}$

Solution:

If $\vec{F}$ **be the resultant force, then** $\overrightarrow{\mathrm{F}}=2\hat{i}+4\hat{j}+2\hat{k}$

Also, $\vec{r}=\overrightarrow{\mathrm{AP}}=-\hat{i}+3\hat{j}+2\hat{k}$

$\therefore$ **required moment** $=\vec{r}\times\vec{F}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-1&3&2\\2&4&2\end{vmatrix}=-2\hat{i}+6\hat{j}-10\hat{k}$

Q.13 Answer:

$0<x<\frac{1}{2}$

Solution:

Angle θ **between vector** $\vec{a}$ **and** $\vec{b}$

$\cos\theta=\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

for obtuse angle,

$\cos\theta<0$

$\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}<0$

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$\vec{a} \cdot \vec{b} < 0$

$14x^2 - 8x + x < 0$

$(7\mathrm{x})(2\mathrm{x} - 1) < 0$

$0 < \mathrm{x} < \frac{1}{2}$

Hence anlge between given vectors are obtuse if $\mathrm{x} \in \left(0, \frac{1}{2}\right)$

Q.14 Answer:

orthocentre of $\triangle \mathrm{ABC}$

Solution:

$\therefore (\vec{a} - \vec{d})(\vec{b} - \vec{c}) = 0$

$\Rightarrow \overrightarrow{\mathrm{DA}}$ **and** $\overrightarrow{\mathrm{CB}}$ **are perpendicular**

$(\vec{b} - \vec{d}) \cdot (\vec{c} - \vec{a}) = 0$

$\Rightarrow \overrightarrow{\mathrm{DB}}$ **and** $\overrightarrow{\mathrm{AC}}$ **are perpendicular**

D **is orthocentre of** $\Delta \mathrm{ABC}$

Q.15 Answer:

circle

Solution:

$(\hat{\mathrm{i}} - 2x\hat{\mathrm{j}} - 3y\hat{\mathrm{k}}) \cdot (\hat{\mathrm{i}} + 3\hat{\mathrm{x}}\hat{\mathrm{j}} + 2\mathrm{y}\hat{\mathrm{k}}) = 0$

$1 - 6\mathrm{x}^2 - 6\mathrm{y}^2 = 0$

$-6\mathrm{x}^2 - 6\mathrm{y}^2 = -1$

$\mathrm{x}^2 + \mathrm{y}^2 = \frac{1}{6}$

$\mathrm{x}^2 + \mathrm{y}^2 = \left(\sqrt{\frac{1}{6}}\right)^2$

Q.16 Answer:

$2\sqrt{5}\hat{i} - 2\sqrt{5}\hat{j} + \sqrt{5}\hat{k}$

Solution:

If $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}$ **and** $\vec{c} = -\hat{i} + 2\hat{k}$

$|\vec{c}| = \sqrt{(-1)^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5}$

$|\vec{c}| \cdot \vec{a} = \sqrt{5} \cdot (2\hat{i} - 2\hat{j} + \hat{k})$

$|\vec{c}| \cdot \vec{a} = 2\sqrt{5}\hat{i} - 2\sqrt{5}\hat{j} + \sqrt{5}\hat{k}$

Q.17 Answer:

50

Solution:

Let $\vec{r}_1 = a\hat{i} + b\hat{j} + c\hat{k}, \vec{r}_2 = 3\hat{i} + 4\hat{j} + 5\hat{k}$

$|\vec{r}_1 \times \vec{r}_2|^2 \leq |\vec{r}_1|^2 |\vec{r}_2|^2$

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Now, $\vec{r}_1 \times \vec{r}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & b & c \\ 3 & 4 & 5 \end{vmatrix}$

$= \hat{i}(5b - 4c) + \hat{j}(3c - 5a) + \hat{k}(4a - 3b)$

So, from (1) **:**

$(5b - 4c)^2 + (3c - 5a)^2 + (4a - 3b)^2 \leq 50$

Q.18 Answer:

$\frac{1}{4}|\mathrm{a} \times \mathrm{b} + \mathrm{b} \times \mathrm{c} + \mathrm{c} \times \mathrm{d} + \mathrm{d} \times \mathrm{a}|$

Solution:

It is given that $\mathrm{a}, \mathrm{b}, \mathrm{c}$, **and** d **are the position vectors of vertices of a quadrilateral** ABCD **respectively.**

Let E, F, G and H **are the middle points of sides** $\mathrm{AB}, \mathrm{BC}, \mathrm{CD}$ **and Da respectively.**

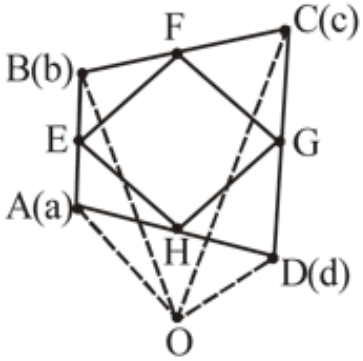
The position vectors of these points will be

$\overrightarrow{\mathrm{OE}} = \frac{1}{2}(\mathrm{a} + \mathrm{b}), \overrightarrow{\mathrm{OF}} = \frac{1}{2}(\mathrm{b} + \mathrm{c})$

$\overrightarrow{\mathrm{OG}} = \frac{1}{2}(\mathrm{c} + \mathrm{d}), \overrightarrow{\mathrm{OH}} = \frac{1}{2}(\mathrm{a} + \mathrm{d})$

Then $\overrightarrow{\mathrm{EF}} = \overrightarrow{\mathrm{OF}} - \overrightarrow{\mathrm{OE}} = \left(\frac{\mathrm{c} - \mathrm{a}}{2}\right)$

and $\overrightarrow{\mathrm{FG}} = 1/2(\mathrm{d} - \mathrm{b}), \overrightarrow{\mathrm{GH}} = 1/2(\mathrm{a} - \mathrm{c}), \overrightarrow{\mathrm{GH}} = 1/2(\mathrm{b} - \mathrm{d})$



It is clear that $\overrightarrow{\mathrm{EF}}$ **is parallel to** $\overrightarrow{\mathrm{GH}}$ **and** $\overrightarrow{\mathrm{FG}}$ **is parallel to** $\overrightarrow{\mathrm{HE}}$**. Thus** EFGH **is a parallelogram.**

$$\begin{aligned} \therefore \overrightarrow{\mathrm{EF}} \times \overrightarrow{\mathrm{FG}} &= 1/4\{(\mathrm{c} - \mathrm{a}) \times (\mathrm{d} - \mathrm{b})\} \\ &= 1/4(\mathrm{c} \times \mathrm{d} - \mathrm{c} \times \mathrm{b} - \mathrm{a} \times \mathrm{d} + \mathrm{a} \times \mathrm{b}) \\ &= 1/4(\mathrm{a} \times \mathrm{b} + \mathrm{b} \times \mathrm{c} + \mathrm{c} \times \mathrm{d} + \mathrm{d} \times \mathrm{a}) \end{aligned}$$

∴ Area of parallelogram EFGH is,

$\mathrm{A} = |\overrightarrow{\mathrm{EF}} \times \overrightarrow{\mathrm{FG}}| = \frac{1}{4}|\mathrm{a} \times \mathrm{b} + \mathrm{b} \times \mathrm{c} + \mathrm{c} \times \mathrm{d} + \mathrm{d} \times \mathrm{a}|$

Q.19 Answer:

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$(3/7)(\mathrm{i}+2\mathrm{j}+3\mathrm{k})$

Solution:

Component of $i+j+k$ along $i+2j+3k$
$= \frac{6}{\sqrt{14}} \times \frac{1}{\sqrt{14}}(i+2j+3k) = \frac{3}{7}(i+2j+3k)$

Q.20 Answer:

$\frac{3}{2}\sqrt{30}$

Solution:

Here $a = 3i+j+k \quad b = i-3j+4k$

$$\text{Thus } \mathrm{a}\times\mathrm{b} = \begin{matrix} i & j & k \\ 3 & 1 & 1 \\ 1 & -3 & 4 \end{matrix}$$

Hence area of parallelogram $= \frac{1}{2}|\mathrm{a}\times\mathrm{b}| = \frac{3}{2}\sqrt{30}$

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